The skeleton of an acyclic cluster variety

Cluster skeleton geometry: formalization reference

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This reference presents the supplied "Skeleton in the acyclic case" argument for the smooth Liouville primitive, complete flow, and explicit dynamical skeleton. The accompanying Lean formalization proves the skew-symmetric unimodular specialization $\widetilde{D} = I$. Weighted notation is retained from the reference argument; a general weighted Lean theorem is not asserted.

1 Skeleton in the acyclic case

Let M and F be the mutable and frozen index sets, and put $n = |M| + |F|$. Fix an integral completion $E \in GL_n(\mathbb{Z})$ whose mutable columns are $\widetilde{B}$. Fix a positive rational diagonal matrix $\widetilde{D} = \mathrm{diag}(d_j)$ such that $K = E\widetilde{D}$ is skew-symmetric. The skew-symmetric case is $\widetilde{D} = I_n$ and $K = E$. The mutable block $B = E_{M,M}$ is acyclic: there is no directed cycle with all successive matrix entries positive. Frozen coordinates remain invertible variables. The unimodular completion hypothesis is retained throughout.

Write $c_i = Ee_i$ and $z^{c_i} = \prod_j z_j^{E_{ji}}$. On the initial torus put

$$\Omega = \frac{1}{2}\sum_{j,l} K_{jl}\, d\log z_j \wedge d\log z_l, \qquad \lambda_{\mathrm{tor}} = u^{\mathsf{T}} K\, d\theta.$$

Here Ω denotes its smooth holomorphic symplectic extension to the exchange variety. The construction and nondegeneracy of that extension, and the smooth atlas on the actual exchange variety, are separate Lean construction obligations; they must not be hidden in the final theorem's hypotheses. The local node charts below give the coordinates for them. For a complete Liouville field, define its dynamical skeleton as the set of points with relatively compact forward orbit. Our sign convention is $\iota_Z d\lambda = \lambda$.

A subset $I \subset M$ is *independent* if $B_{ij} = 0$ for all $i, j \in I$.

In the acyclic case, $\mathcal{A}(\widetilde{B})$ can be presented by generators and relations. For $i \in M$ put

$$P_i^{\pm} = \prod_{j \in M \sqcup F} z_j^{[\pm \widetilde{B}_{ji}]_+}, \qquad t_i = P_i^+ / P_i^- = z^{c_i}.$$

By [1, Corollary 1.21],

$$\mathcal{A}(\widetilde{B}) = \frac{\mathbb{C}[z_i, z_i' \ (i \in M),\ z_j^{\pm 1} \ (j \in F)]}{(z_i z_i' - P_i^+ - P_i^-)_{i \in M}}. \tag{1}$$

Lemma 1. *We have the following.*

(i) *At every point $p \in \mathcal{A}(\widetilde{B})$, the set $\{i \in M \mid z_i(p) = 0\}$ is independent.*

(ii) *Let I be independent, and let $U_I = \{z_j \neq 0 \ (j \in M \setminus I)\} \subset \mathcal{A}(\widetilde{B})$. Then $U_I \cong \mathcal{N}^I \times (\mathbb{C}^*)^{m+f-2|I|}$, where the ith factor $\mathcal{N}$ has coordinates*

$$(x, y, t) = (z_i,\ z_i'/P_i^-,\ t_i) \qquad (i \in I).$$

(iii) *The open sets U_I cover $\mathcal{A}(\widetilde{B})$.*

Proof. (i) Fix a point p, let $Z = \{i \in M : z_i(p) = 0\}$, and suppose that $B_{Z,Z} \neq 0$. Since B is acyclic, we can take a maximal path $j_1, \dots, j_r$ in Z with $r \geq 2$ and $B_{j_s j_{s+1}} > 0$. Put $i = j_r$. By maximality, $B_{ji} \geq 0$ for all $j \in Z$, and $B_{j_{r-1} i} > 0$. So $P_i^+(p) = 0$ since $z_{j_{r-1}}(p) = 0$. Every mutable j in P_i^- has $B_{ji} < 0$, so $j \notin Z$; the frozen z_j are units. So $P_i^-(p) \neq 0$. This contradicts the relation $z_i z_i' = P_i^+ + P_i^-$.

(ii) On U_I, the z_j with $j \in (M \sqcup F) \setminus I$ are units. For $i \in I$, the monomials $P_i^\pm$ involve only these z_j. For $j \in M \setminus I$, we have $z_j' = (P_j^+ + P_j^-)/z_j$. So (1) gives

$$\mathcal{O}(U_I) = \mathbb{C}[z_j^{\pm 1} \ (j \in M \sqcup F \setminus I)]\,[z_i, y_i \ (i \in I)]/(z_i y_i - 1 - t_i)_{i \in I}, \qquad y_i = z_i'/P_i^-.$$

The vectors c_i, $i \in I$, are columns of $E \in GL_{m+f}(\mathbb{Z})$ and lie in $\mathbb{Z}^{(M \sqcup F) \setminus I}$. So they extend to a basis of $\mathbb{Z}^{(M \sqcup F) \setminus I}$, which implies $\mathbb{C}[z_j^{\pm 1} \ (j \notin I)] = \mathbb{C}[t_i^{\pm 1} \ (i \in I)] \otimes \mathcal{O}((\mathbb{C}^*)^{m+f-2|I|})$. This proves (ii).

(iii) For any point $p \in \mathcal{A}(\widetilde{B})$, take $I = \{i \in M \mid z_i(p) = 0\}$. Then $p \in U_I$. □

Cutoffs and the local node

The node is $\mathcal{N} = \{(x, y, t) \in \mathbb{C}^2 \times \mathbb{C}^* : xy = 1 + t\}$. On $x \neq 0$ write $x = e^{u+i\theta}$ and $t = e^{v+i\eta}$. Its holomorphic symplectic form is $d\log t \wedge d\log x = dy \wedge dx/(xy-1)$. Choose $L > a > 0$ and smooth functions satisfying

$$\begin{gathered}
h \leq 0, \quad 0 \leq h' \leq 1, \quad u - h(u) < 0 \ (u < 0), \\
h(u) = u + a \ (u \leq -L), \qquad h = 0 \text{ near } [0, \infty), \\
\rho \in C_c^\infty(\mathbb{R}, [0,1]), \qquad \rho = 1 \text{ near } 0, \\
\beta \in C^\infty(\mathbb{R}/2\pi\mathbb{Z}), \quad \operatorname{supp}\beta \subset (\pi - \varepsilon, \pi + \varepsilon), \\
\beta(\eta) = \eta - \pi \text{ near } \pi, \quad (\eta - \pi)\beta(\eta) \geq 0, \quad \beta' \leq 1, \qquad 0 < \varepsilon < \pi.
\end{gathered}$$

The angular difference is the lift in $(-\varepsilon, \varepsilon)$ on this arc. The node primitive is $v\,d\theta - u\,d\eta + d(h\rho\beta)$. These cutoff properties, including existence, are part of the formalization.

Theorem 1. *There is a Liouville form λ_{ac} of $\Im\Omega$ on $\mathcal{A}(\widetilde{B})$ such that*

$$\begin{gathered}
\Lambda = \operatorname{Skel}(Z_{\lambda_{\mathrm{ac}}}) = \bigcup_{I \subset M \text{ independent}} \Lambda_I, \\
\Lambda_I = \{z_i' = 0, \ |z_i| \leq 1 \ (i \in I), \quad |z_j| = 1 \ (j \in (M \sqcup F) \setminus I)\}.
\end{gathered} \tag{2}$$

Proof. Use the constants $L > a > 0$ and the cutoffs h, ρ, β specified above. Let $\lambda_{\mathrm{tor}} = u^{\mathsf{T}} K \mathrm{d}\theta$. Fix $\delta_0 > 0$ with $h = 0$ on $(-\delta_0, \infty)$. Choose a smooth nonincreasing $\vartheta\colon \mathbb{R} \to [0,1]$ with $\vartheta = 1$ on $(-\infty, -1]$ and $\vartheta = 0$ on $[0, \infty)$. On the initial torus $(\mathbb{C}^*)^{m+f}$ write

$$u_j = \log|z_j|, \qquad \theta_j = \arg z_j, \qquad v_i = \log|t_i| = c_i(u), \qquad \eta_i = \arg t_i = c_i(\theta),$$

and set

$$\chi_i := \prod_{j \in M:\, B_{ji} \neq 0} \vartheta(u_i - u_j), \qquad H := \sum_{i \in M} d_i\, h(u_i)\, \chi_i\, \rho(v_i)\, \beta(\eta_i), \qquad \lambda_{\mathrm{ac}} := \lambda_{\mathrm{tor}} + \mathrm{d}H.$$

An empty product is 1. Note that if $B_{ij} \neq 0$, then $\chi_i \chi_j = 0$.

Let $I \subset M$ be independent. For $i, i' \in I$, $K_{ii'} = d_{i'} B_{ii'} = 0$. For $i \in M$,

$$\sum_j K_{ji}\, u_j = d_i \sum_j E_{ji} u_j = d_i v_i, \qquad \sum_l K_{il}\, \mathrm{d}\theta_l = -\sum_l K_{li}\, \mathrm{d}\theta_l = -d_i\, \mathrm{d}\eta_i.$$

The terms of λ_{tor} with $l \in I$ give $\sum_{i\in I} d_i v_i \,\mathrm{d}\theta_i$. The terms with $j \in I$ and $l \notin I$ give $-\sum_{i\in I} d_i u_i \,\mathrm{d}\eta_i$, since $K_{il} = 0$ for $i, l \in I$. Hence

$$\lambda_{\mathrm{tor}} = \sum_{i\in I} d_i(v_i \,\mathrm{d}\theta_i - u_i \,\mathrm{d}\eta_i) + \sum_{j,l\in(M\sqcup F)\setminus I} K_{jl}\, u_j \,\mathrm{d}\theta_l. \tag{3}$$

We first show that λ_{ac} is globally defined. Let $q \in \mathcal{A}(\widetilde{B})\setminus(\mathbb{C}^*)^{m+f}$, and let $I = \{i \in M \mid z_i(q) = 0\}$. Then $I \neq \emptyset$, I is independent by Lemma 1(i), and $q \in U_I$. Let $i \in I$. The monomials $P_i^{\pm}$ only involve z_j with $j \notin I$, so they are units on U_I. Hence $t_i = P_i^+/P_i^-$ is a unit on U_I, and $0 = z_i(q)z_i'(q) = P_i^-(q)(1 + t_i(q))$ gives $t_i(q) = -1$. So, as $p \to q$ with $p \in (\mathbb{C}^*)^{m+f}$,

$$u_i(p) \to -\infty, \qquad v_i(p) \to 0, \qquad \eta_i(p) \to \pi, \qquad u_j(p) \to \log|z_j(q)| \in \mathbb{R} \quad (j \in (M \sqcup F) \setminus I).$$

Choose a neighborhood $W \subset U_I$ of q such that, for $i \in I$ and $j \in M$ with $B_{ji} \neq 0$,

$$\rho(v_i) = 1, \quad \beta(\eta_i) = \eta_i - \pi \quad \text{on } W, \qquad u_i \le -L, \quad u_i \le u_j - 1 \quad \text{on } W \cap (\mathbb{C}^*)^{m+f}.$$

Then, on $W \cap (\mathbb{C}^*)^{m+f}$, $h(u_i) = u_i + a$ and $\chi_i = 1$. Also $\chi_j = 0$ there, because χ_j has the factor $\vartheta(u_j - u_i)$ and $u_j - u_i \ge 1$. So, on $W \cap (\mathbb{C}^*)^{m+f}$,

$$H = \sum_{i\in I} d_i(u_i + a)(\eta_i - \pi) + H', \tag{4}$$

where H' is the sum of the summands of H with index $k \in M \setminus I$ such that $B_{ik} = 0$ for all $i \in I$. For such k, the functions u_k, χ_k, $v_k = c_k(u)$ and $\eta_k = c_k(\theta)$ only involve z_j with $j \notin I$, by the support of c_k. These z_j are units on U_I. So H' does not depend on u_i and θ_i, $i \in I$.
By (3) and (4), on $W \cap (\mathbb{C}^*)^{m+f}$,

$$\lambda_{\mathrm{ac}} = \sum_{j,l\notin I} K_{jl}\, u_j \,\mathrm{d}\theta_l + \mathrm{d}H' + \sum_{i\in I} d_i\big(v_i \,\mathrm{d}\theta_i + (\eta_i - \pi)\,\mathrm{d}u_i + a\,\mathrm{d}\eta_i\big).$$

Here we used $v_i \,\mathrm{d}\theta_i - u_i \,\mathrm{d}\eta_i + \mathrm{d}\big((u_i + a)(\eta_i - \pi)\big) = v_i \,\mathrm{d}\theta_i + (\eta_i - \pi)\,\mathrm{d}u_i + a\,\mathrm{d}\eta_i$. The first two terms are smooth on W. On W, $1 - z_iy_i = -t_i$ and $\mathrm{Log}(-t_i) = v_i + \mathrm{i}(\eta_i - \pi)$. So, by the holomorphic logarithm near 1,

$$v_i \,\mathrm{d}\theta_i + (\eta_i - \pi)\,\mathrm{d}u_i + a\,\mathrm{d}\eta_i = \Im\Big(\frac{\mathrm{Log}(1 - z_iy_i)}{z_i}\,\mathrm{d}z_i\Big) + a\,\mathrm{d}\arg(1 - z_iy_i).$$

The right side is smooth on W, because $\mathrm{Log}(1 - z_iy_i)/z_i$ is holomorphic on W, with value $-y_i$ on $\{z_i = 0\}$. Hence λ_{ac} extends smoothly over W, hence also extends to $\mathcal{A}(\widetilde{B})$.
Now we consider the Liouville field $Z = Z_{\lambda_{\mathrm{ac}}}$, given by $\iota_Z \Im\Omega = \lambda_{\mathrm{ac}}$. A dot denotes the derivative along Z. On the initial torus, $\Im\Omega = \sum_{j,l} K_{jl}\,\mathrm{d}u_j \wedge \mathrm{d}\theta_l$. Comparing the coefficients of $\mathrm{d}\theta$ and of $\mathrm{d}u$ in $\iota_Z\Im\Omega = \lambda_{\mathrm{tor}} + \mathrm{d}H$ gives

$$K^{\mathsf{T}}\dot{u} = K^{\mathsf{T}}u + \partial_\theta H, \qquad -K\dot{\theta} = \partial_u H.$$

The function H depends on θ only through $\eta_i = c_i(\theta)$, and $d_ic_i = Ke_i$. So

$$\partial_\theta H = \sum_{i\in M} d_i\, h(u_i)\chi_i\rho(v_i)\beta'(\eta_i)\, c_i = K\sum_{i\in M} h(u_i)\chi_i\rho(v_i)\beta'(\eta_i)\, e_i.$$

Since $K^{\mathsf{T}} = -K$, this gives

$$\begin{aligned} \dot{u}_j &= u_j && (j \in F),\\ \dot{u}_i &= u_i - h(u_i)\,\chi_i\,\rho(v_i)\,\beta'(\eta_i) && (i \in M),\\ \dot{\theta} &= -K^{-1}\partial_u H. \end{aligned} \tag{5}$$

For $i \in M$, $c_i(K^{-1}x) = -x_i/d_i$, because $d_i c_i^{\mathsf{T}} = e_i^{\mathsf{T}} K^{\mathsf{T}} = -e_i^{\mathsf{T}} K$. Also $c_i(e_k) = E_{ki} = B_{ki}$ for $k \in M$. So applying c_i to (5) gives

$$\dot\eta_i = \frac{1}{d_i}\,\partial_{u_i} H, \qquad \dot v_i = v_i - \sum_{k\in M} B_{ki}\, h(u_k)\,\chi_k\,\rho(v_k)\,\beta'(\eta_k) \qquad (i \in M). \tag{6}$$

We now bound $\dot u$. Since $h \le 0$, $0 \le \chi_i\rho \le 1$ and $\beta' \le 1$, we have $h\chi_i\rho\beta' \ge h\chi_i\rho \ge h$. With $h = 0$ on $(-\delta_0, \infty)$, (5) gives, for $i \in M$,

$$\dot u_i \le u_i - h(u_i) \quad \text{on } \{u_i < 0\}, \qquad \dot u_i = u_i \quad \text{on } \{u_i > -\delta_0\}. \tag{7}$$

Recall that $u - h(u) < 0$ for $u < 0$, and $u - h(u) = -a$ for $u \le -L$. Also $|h(u)| \le |u| + L$, since $h(u) = u + a$ for $u \le -L$, and $a - L \le h \le 0$ on $[-L, \infty)$. So (5) gives $|\dot u_j| \le C(1 + |u_j|)$ for all $j \in M \sqcup F$, with $C = (1 + \max|\beta'|)(1 + L)$. By density, (7) holds on $\{z_i \ne 0\}$, and the bound holds on $\{z_j \ne 0\}$.

Next we compute Z near $\{z_i = 0\}$. Let q, I and W be as above, and let $i \in I$. On $W \cap (\mathbb{C}^*)^{m+f}$, $h(u_i) = u_i + a$, $\chi_i = \rho(v_i) = \beta'(\eta_i) = 1$, and $\chi_k = 0$ for $B_{ki} \ne 0$. Also $\partial_{u_i} H = d_i(\eta_i - \pi)$ there, by (4). So (5) and (6) give

$$\dot u_i = -a, \qquad \dot v_i = v_i, \qquad \dot\eta_i = \eta_i - \pi \qquad \text{on } W \cap (\mathbb{C}^*)^{m+f}.$$

Since $\operatorname{Log}(-t_i) = v_i + \mathrm{i}(\eta_i - \pi)$, by density

$$Z|z_i|^2 = -2a|z_i|^2, \qquad \dot t_i = t_i \operatorname{Log}(-t_i) \qquad \text{on } W. \tag{8}$$

Every point of $\{z_i = 0\}$ is such a q, so these sets W cover $\{z_i = 0\}$. Along an orbit near $\{z_i = 0\}$, $|z_i|^2$ solves $\frac{\mathrm{d}}{\mathrm{d}\tau}|z_i|^2 = -2a|z_i|^2$. By uniqueness of solutions, $\{z_i = 0\}$ and its complement are invariant under the flow of Z.

On $\{z_i = 0\}$, the z_j with $B_{ji} \ne 0$ are nonzero, by Lemma 1(i), so $P_i^- \ne 0$. So $y_i = z_i'/P_i^-$ is defined near $\{z_i = 0\}$, and $y_i = (1 + t_i)/z_i$ where $z_i \ne 0$. Put $\phi(t) = t\operatorname{Log}(-t)/(1 + t)$. It is holomorphic near $t = -1$, with $\phi(-1) = 1$. On $W \cap \{z_i y_i \ne 0\}$, (8) gives

$$Z \log|y_i| = \Re\frac{\dot t_i}{1 + t_i} - Z\log|z_i| = \Re\phi(t_i) + a.$$

By density, $Z|y_i|^2 = 2(\Re\phi(t_i) + a)|y_i|^2$ on W. On $W \cap \{z_i = 0\}$, $t_i = -1$, so

$$Z|y_i|^2 = 2(1 + a)|y_i|^2 \qquad \text{on } \{z_i = 0\}. \tag{9}$$

Hence $|y_i(\phi_Z^\tau(p))| = e^{(1+a)\tau}|y_i(p)|$ for $p \in \{z_i = 0\}$ and all τ.

Next we show that Z is complete. Let $\gamma\colon (\tau_-, \tau_+) \to \mathcal{A}(\widetilde B)$ be a maximal integral curve of Z, and let $I_0 = \{i \in M \mid z_i(\gamma(0)) = 0\}$. By invariance, $z_i = 0$ on γ for $i \in I_0$, and $z_j \ne 0$ on γ for $j \in (M \sqcup F) \setminus I_0$. Let $T > 0$. For $\tau \in (\tau_-, \tau_+)$ with $|\tau| \le T$, the bound $|\dot u_j| \le C(1 + |u_j|)$ and (9) give

$$|u_j(\gamma(\tau))| \le (1 + |u_j(\gamma(0))|)e^{CT} \quad (j \notin I_0), \qquad |y_i(\gamma(\tau))| \le e^{(1+a)T}|y_i(\gamma(0))| \quad (i \in I_0).$$

So the coordinates $z_j^{\pm 1}$ $(j \notin I_0)$, $z_j' = (P_j^+ + P_j^-)/z_j$ $(j \in M \setminus I_0)$ and $z_i' = P_i^- y_i$ $(i \in I_0)$ are bounded on these $\gamma(\tau)$. By (1), $\mathcal{A}(\widetilde B)$ is closed in $\mathbb{C}^{2m} \times (\mathbb{C}^*)^f$, so these $\gamma(\tau)$ lie in a compact set. Hence $\tau_\pm = \pm\infty$, since a maximal integral curve with $\tau_+ < \infty$ or $\tau_- > -\infty$ leaves every compact set.

We now prove (2).

(i) $\operatorname{Skel}(Z) \subset \cup_I \Lambda_I$. Let $p \in \operatorname{Skel}(Z)$, so that the closure of $\gamma(\tau) = \phi_Z^\tau(p)$, $\tau \ge 0$, is compact. Put

$$I_0 = \{i \in M \mid z_i(p) = 0\}, \qquad I = I_0 \cup \{i \in M \mid z_i(p) \ne 0,\ u_i(p) < 0\}.$$

We show that $p \in \Lambda_I$.

We first determine u along γ. For $j \in F$, $u_j(\gamma(\tau)) = e^\tau u_j(p)$ is bounded, so $u_j(p) = 0$. Let $i \in M \setminus I_0$. If $u_i(p) > 0$, then $u_i(\gamma(\tau)) = e^\tau u_i(p)$ by (7), which is unbounded. If $u_i(p) = 0$, then $u_i = 0$ on γ, by (7) and uniqueness for $\dot{u}_i = u_i$. If $u_i(p) < 0$, then u_i decreases along γ, and (7) gives

$$\dot{u}_i \le \sup_{u \le u_i(p)} (u - h(u)) < 0 \qquad \text{on } \gamma.$$

The supremum is negative because $u - h(u) < 0$ for $u < 0$ and $u - h(u) = -a$ for $u \le -L$. So $u_i(\gamma(\tau)) \to -\infty$. In summary,

$$u_j = 0 \text{ on } \gamma \quad (j \in (M \sqcup F) \setminus I), \qquad u_i(\gamma(\tau)) \to -\infty \quad (i \in I \setminus I_0), \qquad z_i = 0 \text{ on } \gamma \quad (i \in I_0).$$

Let q be a limit point of γ. Then $\{i \in M \mid z_i(q) = 0\} = I$, so I is independent by Lemma 1(i). Also $|z_j| = 1$ on γ and at q for $j \in M \setminus I$. So the closure of γ is a compact subset of U_I. On U_I, $z_i' = P_i^- y_i$ with P_i^- a unit, for $i \in I$. Since $|z_i(p)| \le 1$ for $i \in I$ and $|z_j(p)| = 1$ for $j \notin I$, it remains to show that $y_i(p) = 0$ for $i \in I$.

Let $i \in I_0$. By (9), $|y_i(\gamma(\tau))| = e^{(1+a)\tau}|y_i(p)|$, which is bounded only if $y_i(p) = 0$.

Let $i \in I \setminus I_0$. We show that $t_i = -1$ on γ. The character c_i is supported on $F \sqcup \{j \in M \mid B_{ji} \neq 0\}$, which is disjoint from I. So $v_i = c_i(u) = 0$ on γ. Consider the open set

$$O = \{z_i \neq 0,\ |z_j| > e^{-\delta_0}\ (j \in M,\ B_{ji} \neq 0)\} \supset \gamma.$$

Besides the ith summand of H, only the summands with index j, $B_{ji} \neq 0$, depend on u_i, through χ_j and $v_j = c_j(u)$. These vanish on $O \cap (\mathbb{C}^*)^{m+f}$, because $h(u_j) = 0$ there. Also $\partial_{u_i} v_i = E_{ii} = 0$. So (6) gives, on $O \cap (\mathbb{C}^*)^{m+f}$, and then on O by density,

$$\dot{\eta}_i = \big(h'(u_i)\,\chi_i + h(u_i)\,\partial_{u_i}\chi_i\big)\,\rho(v_i)\,\beta(\eta_i).$$

The factor in front of $\beta(\eta_i)$ is ≥ 0, because $h' \ge 0$, $h \le 0$, $\chi_i \ge 0$, $\rho \ge 0$, and $\partial_{u_i}\chi_i \le 0$ as $\vartheta' \le 0$. Since $(\eta - \pi)\beta(\eta) \ge 0$ and $\operatorname{supp}\beta \subset (\pi - \varepsilon, \pi + \varepsilon)$, the distance $|\eta_i - \pi|$ in $\mathbb{R}/2\pi\mathbb{Z}$ is nondecreasing along γ. On the other hand, $z_i(\gamma(\tau)) \to 0$, and y_i is bounded on γ. So $1 + t_i = z_i y_i \to 0$, and $|\eta_i - \pi| \to 0$. Hence $\eta_i = \pi$ on γ. With $v_i = 0$, this gives $t_i = -1$ on γ, so $y_i = (1 + t_i)/z_i = 0$ on γ. Thus $p \in \Lambda_I$, and $\operatorname{Skel}(Z) \subset \bigcup_I \Lambda_I$.

(ii) $\Lambda_I \subset \operatorname{Skel}(Z)$.

Let I be independent, and let $p \in \Lambda_I$. The set Λ_I is closed, and on it the coordinates $z_j^{\pm 1}$ $(j \notin I)$, z_i, $z_i' = 0$ $(i \in I)$ and $z_j' = (P_j^+ + P_j^-)/z_j$ $(j \in M \setminus I)$ are bounded. So Λ_I is compact, and it suffices to show $\phi_Z^\tau(p) \in \Lambda_I$ for $\tau \ge 0$. Let $\gamma(\tau) = \phi_Z^\tau(p)$. As above, $u_j = 0$ on γ for $j \notin I$. For $i \in I$, $z_i = 0$ on γ if $z_i(p) = 0$, and $u_i \le 0$ on γ if $z_i(p) \neq 0$. It remains to show $y_i = 0$ on γ for $i \in I$. If $z_i(p) = 0$, this follows from (9) and $y_i(p) = 0$. If $z_i(p) \neq 0$, then $v_i = 0$ on γ, and η_i solves the equation for $\dot{\eta}_i$ above, with $\eta_i(p) = \pi$. Since $\beta(\pi) = 0$, uniqueness gives $\eta_i = \pi$ on γ. So $t_i = -1$ and $y_i = 0$ on γ. This proves (2). □

Formalization target and convention matching

All geometry is on the actual complex points of the exchange equations with their induced topology. Construct the smooth presentation, the extended symplectic form, the primitive, and a jointly smooth flow defined for every real time. Define the coordinate cap and dynamical trapped set independently, and prove their equality for this same flow. The equivalent coordinate cap is

$$\{p : |z_f| = 1\ (f \in F),\ |z_i| \le 1,\ (1 - |z_i|^2)z_i' = 0\ (i \in M)\}.$$

For the converse equivalence, the indices with $|z_i| < 1$ have $z_i' = 0$; the zero-set argument applied to their exchange relations makes them independent (equivalently use the existing strict-cap-independence lemma).

State of the formalization. Both descriptions are now formalized, and (2) holds literally in Lean for the constructed flow. The stratum Λ_I of (2) is `ClusterSkeleton.lambdaStratum` in `ClusterSkeleton/CapStrata.lean`, which also proves `lambdaStratum_subset_coordinateCap`, `coordinateCap_subset_lambdaStratum_strictCapIndices`, `coordinateCap_eq_iUnion_lambdaStratum` (the union over the subtype of independent I), `isClosed_lambdaStratum` and `isCompact_lambdaStratum` — the per-stratum compactness used in inclusion (ii) above. Composed with the cap equality, this yields `ClusterSkeleton.FullSkeleton.trappedSet_eq_iUnion_lambdaStratum`, which is (2) for the same constructed flow. Independence of I is used for neither inclusion; it only indexes the union by the family used here.

The torus exponent $m + f - 2|I|$ of Lemma 1(ii) is `ClusterSkeleton.IndependentNodeChart.complementRank_eq_sub_two_mul_card`, namely `complementRank(independentRowRightInverseSI)=d-2*I.card` for independent I in the skew-symmetric specialization $d = m + f$, with `two_mul_card_le` for $2|I| \leq d$. Both are in `ClusterSkeleton/IndependentNodeSmooth.lean`, in namespace `ClusterSkeleton.IndependentNodeChart`.

These are restatements of the proved theorem and of the existing chart cardinality lemma: no hypothesis is added, the source hypotheses are unchanged, and the scope stays the skew-symmetric unimodular case with mutable acyclicity and variable frozen coordinates. The existing Lean seed uses row exponents, whereas this reference uses column exponents. In the skew-symmetric specialization the two exchange monomials are interchanged, so their sum and the exchange variety agree. Exchange ratios are inverted. The sign of the logarithmic form and the angle-flow conventions are compared in the accompanying proof-to-code audit.

No principal-coefficient assumption is imposed. Unimodularity is retained; neither removal of acyclicity nor arbitrary singular matrices is asserted. The formalization uses native Mathlib geometry and the pinned FsFltz interfaces. The verification records bind the accepted source to its exact theorem types and transitive axiom audit; the permitted logical foundations are `propext`, `Classical.choice`, and `Quot.sound`.

References

[1] A. Berenstein, S. Fomin, A. Zelevinsky, *Cluster algebras III: Upper bounds and double Bruhat cells*, Duke Math. J. 126 (2005), 1–52, Corollary 1.21.